Omnikart: an e-shopkeeper you can talk to

Megathon 2026, Accessible AI for All, shopping track. Deliverables: solution, approach, features and results.

1. Solution

Online shopping assumes you can see the page. A blind shopper using a screen reader has to listen to the whole shop: dozens of filters, a carousel and a long specification sheet, before reaching a single price. Older shoppers and people with limited hand movement face the same wall in a different form.

Omnikart is a voice guide that sits on top of an ordinary online store and does what a good shopkeeper does. It asks what you want, asks a few short questions to narrow the choice, suggests the best three, reads what other buyers said, tells you when it will arrive, and places the order only when you clearly say so. It works in English, Hindi and Telugu.

The store and the guide are two separate pieces. Our demo store, Haat, looks and behaves like a familiar shopping site with 60 products in 5 categories. Omnikart talks to it only through a small published contract, so the same guide can be attached to any store that answers the same questions.

2. Approach

- The AI decides what to do; code decides what is true. The language model picks one action from a fixed list. Every price, rating, date and total is read from the store's data by plain code. The model never writes a number.
- Money moves only on a clear yes. Omnikart reads back the items, address, delivery date, payment and the total twice. Only after that read-back has been heard to the end does it accept the exact phrase "yes, place order". A bare "yes", silence or an interruption does nothing. The store checks again and refuses any order that arrives without a confirmation.
- Preferences, never diagnoses. On the first visit Omnikart introduces itself and asks, by voice, which language you want and how you like to be spoken to. It does not ask what disability you have. It also notices signals such as repeated mis-taps or "what?" and offers one helpful change, once.
- No dead ends. With every search the store reports how many products each filter option would leave. Omnikart reads only the options that lead somewhere and skips a question that cannot narrow anything.
- Fail gently. If a request fails or is outside the demo, Omnikart says so, puts everything back as it was, and repeats its last announcement. Nothing restarts. If the connection drops it says so and retries when it returns.
- Built to plug in. The store and the guide share only a contract of twelve calls and a few page events. A lint rule fails the build if either side imports the other.
- Careful with free services. Speech recognition, translation and text to speech use AnuvaadHub; the language model is Groq's free tier. Keys are pooled and each key is rested before it reaches

its per-minute limit. Short answers to filter questions are understood without any translation or AI call.

3. Features

- Spoken welcome and settings by voice, language first, then speech speed, larger text and stronger colours, one fact at a time, one-key mode and screen-reader mode. Every question also has large buttons.
- Guided six-step journey: ask, filters, best matches, reviews, delivery, order.
- Filter answers understood in English, Hindi and Telugu, including spoken numbers such as "under fifteen thousand".
- The store page follows the conversation: filter boxes tick themselves as the shopper speaks, and a sighted helper clicking the page changes Omnikart's next answer.
- Best three matches, each with an honest reason, and how many were left out.
- Reviews in short: notes from shoppers with disabilities first, then what people liked, what they did not, and a verdict.
- Delivery timeline: dispatch day, arrival date, free delivery, return window and seller.
- Safe checkout with full read-back, the total spoken twice, and a labelled button that stays locked until the read-back is finished.
- One-key mode for limited hand movement: Omnikart reads the choices and Space selects.
- Calm mode for older shoppers: slower speech, one fact at a time.
- High contrast, larger text, captions always on, and a screen edge that flashes with every sound so nothing depends on hearing alone.
- Prices spoken as whole numbers in all three languages.
- Control room: a popup showing raw page content beside what Omnikart says, a log of every API call, the decision log in plain words, and switches to make each service fail on purpose.
- The store runs entirely on its own with the guide switched off.

4. Results

What we have verified:

- 384 automated tests pass. Type checking, lint (including the rule that keeps store and guide apart) and the production build are clean.
- The full ten-turn purchase runs end to end over the store's HTTP API with typed input.
- Independence, both ways. The store alone completes a purchase by clicking and shows zero axe accessibility violations on all six of its pages. The guide completes the whole journey against a made-up store with different products, filters, addresses and page addresses, with no change to its code.
- Checked in a real browser: the first-visit welcome and settings questions, switching to Hindi by answer, the store page following spoken filters, the order button staying locked until the read-

back ends, the order announcement, the fallback message, and the Control room popup with its call log.

- Guard rails are tested: a bare "yes" places nothing, an interrupted read-back cancels, an AI rewrite that changes a number or a name is discarded, and the store refuses an unconfirmed order.

What we have not yet verified, stated plainly:

- The spoken path with live speech services across a full session. Our browser runs used typed input with the speech services switched off.
- Testing with blind, low-vision, older and motor-impaired shoppers. This is the most important next step.
- The Hindi and Telugu wording, which still needs a native speaker's review.

Assumptions and declarations:

- Haat is our own demo store. Brand names are real and used for realism; models, specifications, prices, sellers and reviews are illustrative and are not real offers. Reviews are synthetic and generated from a fixed seed. Product pictures are our own drawings.
- No real payment is taken. The delivery address is a demo address.
- The guide needs a store that implements the contract. It does not read arbitrary websites.
- Browsers allow sound only after a first key press or touch, so the welcome begins on that first touch.
- AI rewording of sentences is used for English only; Hindi and Telugu use sentences we wrote ourselves.

5. Future scope

- Trials with real shoppers, and changes driven by what they tell us.
- More Indian languages as speech models allow.
- A phone-call version for people without a smartphone.
- Connecting real stores through the same contract.
- Photo and size-chart description for stores whose images carry no text.

6. Potential business model

Omnikart is a plug-in for online stores, not a store. A shop implements the contract once and pays per order completed through the guide, or a flat monthly fee. The shopper never pays. For the store it opens a customer group it cannot serve well today and helps meet accessibility obligations; for the shopper it is simply a shop that listens. This model is a proposal: billing is not built.

Accessibility should not be a feature added at the end. It should be the front door.